

Live Causal Navigator

Pneumococcal vaccination and pneumonia

Module F · Standalone workshop

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MODULE D → MODULE F

Same causal logic.
Every decision
becomes visible,
revisable, and
exportable.

Workshop stance

Navigator records choices;
it is not an oracle.

CAUSAL INFERENCE FOR REAL-WORLD DATA

A Causal Navigator

Causal inference requires three ingredients.
Our tools help you formalize each one rigorously.

We endorse the **question-first** approach

We align with the philosophy of **Targeted Learning** as articulated by Mark van der Laan: start with what you're trying to learn from the data, not with convenient statistical activities

Question-First

- Q Start with your causal question
- D Assess your available data
- M Formalize your assumptions

Know if your question is answerable before you analyze

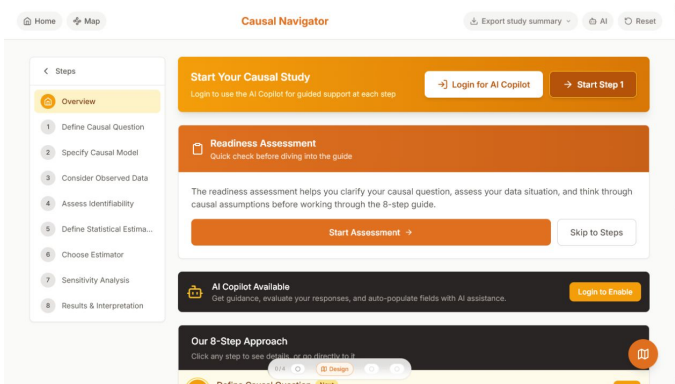
× NOT activity-first

1. Start with data
2. Run statistical tests
3. Hope results are meaningful

Risk of answering the wrong question or finding spurious associations

When the three ingredients are specified, your answer will have a **causal interpretation**

Q D M Causal



- **Start at Step 1.** Use the manual route.
- **Work in order.** Pause at design decisions.
- **Export at Step 8.** Keep the Markdown audit trail.

AI support is optional and is not used in this workshop.

navigator.tao-rwd.com

Observed 12-month pneumonia risk

7.19%

vaccinated

6.43%

unvaccinated

Crude RD **+0.76 pp**

Individual prediction → pair check

Name the population, time zero, follow-up, and contrast needed to interpret the difference.

Higher-risk adults are more likely to be vaccinated. The association mixes baseline risk with the treatment effect.

Population	Synthetic adult cohort, $N = 5,000$; age centered at 65 (not restricted to age 65+)
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Treatment	Vaccination at baseline ($A = 1$)
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Comparator	No vaccination at baseline ($A = 0$)
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Outcome	Pneumonia hospitalization within 12 months
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Contrast	Average 12-month risk difference
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Time-zero rule

Eligibility, treatment assignment, and follow-up all begin at the same time zero.

1 Define the causal question

- intervention, comparator, and time zero
- population and eligibility
- outcome horizon
- causal contrast

Checkpoint

Which phrase most changes the estimand?

The screenshot shows the 'Causal Navigator' web application. At the top, there are navigation links for 'Home' and 'Map', and a title 'Causal Navigator'. On the right, there are buttons for 'Export study summary', 'AI', and 'Reset'. A left sidebar titled 'Steps' contains a list of steps: 'Overview', '1 Define Causal Question' (highlighted), '2 Specify Causal Model', '3 Consider Observed Data', '4 Assess Identifiability', '5 Define Statistical Estima...', '6 Choose Estimator', '7 Sensitivity Analysis', and '8 Results & Interpretation'. The main content area is divided into three sections: 'Outcome' with the text 'Hospitalization with pneumonia within 12 months of time zero (binary Y).', 'Follow-up Period' with the text 'Start at the aligned index date (vaccination or eligible non-vaccination date) and end at 365 days. Censor at death, disenrollment, loss to follow-up, or database end. The teaching simulation has complete 1-year outcomes.', and 'Summary Measure (Causal Contrast)' with the text 'Primary: 1-year risk difference, $E[Y^*1] - E[Y^*0]$. Secondary: 1-year risk ratio, $E[Y^*1] / E[Y^*0]$.' At the bottom, there is a progress bar showing '0/4' steps, with 'Design' highlighted, and a 'Step 2' button.

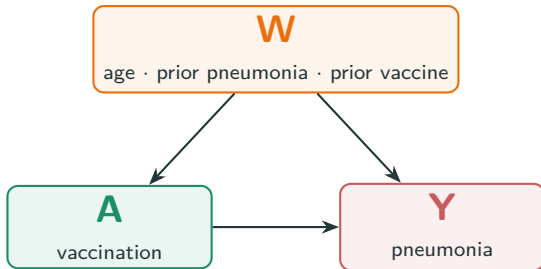
THE WORKSHOP VIEW

A compact model makes the confounding structure visible.

- Stipulated baseline set: age, prior pneumonia, and prior vaccination.
- Do not adjust for post-baseline descendants.
- The kit contains the full five-node, nine-edge DAG.

Accept, correct, or reject?

Does this adjustment set follow from the DAG you are willing to defend?



Collapsed discussion view; use the complete DAG for the audit.

3 Map the
data
5,000 people
1,627
vaccinated
334 events

One row per person;
complete baseline
covariates, treatment,
and 12-month outcome.

Protocol	Observed-data mapping
Population	One row per eligible participant
Treatment	Baseline vaccination indicator A
Outcome	12-month pneumonia hospitalization Y
Baseline W	Age, prior pneumonia, prior vaccination
Time zero	Eligibility, treatment, and follow-up aligned
<i>Mapping check: Can each protocol element be represented without changing the question?</i>	

Can the observed data answer the causal question?

- exchangeability given W
- practical positivity
- consistency
- no interference

Three-part audit

Choose the weakest assumption, evidence you would seek, and one design change if it fails.

Assumption	Data can inform	Requires judgment
Exchangeability	Balance; proxies	Are key causes recorded?
Positivity	Fitted scores; support	Is each target stratum supported?
Consistency	Recorded versions	Is the intervention well defined?
No interference	Cluster/context clues	Can one person's exposure affect another?

The generator states the assumptions for this exercise; observed data alone cannot certify them.

CAUSAL TARGET

$$\psi = E[Y^1] - E[Y^0]$$

OBSERVED-DATA MAPPING

$$E_W[E(Y \mid A = 1, W) - E(Y \mid A = 0, W)]$$

Audit question

What changes if the population, outcome window, or censoring changes?

The screenshot displays the 'Causal Navigator' web application. On the left, a sidebar lists the steps: Overview, Define Causal Question, Specify Causal Model, Consider Observed Data, Assess Identifiability, Define Statistical Estimand (highlighted), Choose Estimator, Sensitivity Analysis, and Results & Interpretation. The main panel shows the 'Define Statistical Estimand' step with three sections: 'Statistical Estimand' containing the formula $\text{psi_RD} = E_W\{E(Y \mid A=1, W) - E(Y \mid A=0, W)\}$, 'Link to Causal Under Identification' explaining the conditions for the statistical estimand to equal the causal effect, and 'Censoring Adjustment' stating that no adjustment is needed for the teaching dataset but it should be modeled in a real study. The bottom of the interface shows 'Step 4' and a 'Design' button.

6 Choose the estimator

- **Primary:** TMLE for the risk difference
- $\bar{Q}(A, W)$: outcome regression
- $g(W)$: treatment mechanism
- influence-function 95% CI

Boundary question

What can an estimator repair—and what can it never repair?

Pre-result workflow

No repeated simulation or bias/RMSE study is included. The adjusted outcome stays hidden until Steps 1–7 are locked.

Overlap diagnostic

Fitted propensity scores range from 0.0765 to 0.9091. This diagnoses overlap; it does not prove structural positivity.

Reproducibility convention

Report the version-sensitive TMLE result as -3.0 points (95% CI $-4.6, -1.5$).

Prespecify stress tests before results.

- unmeasured confounding
- outcome misclassification
- selection and censoring
- alternative learners and truncation

Choose one gap

What analysis or evidence would clarify it?

Sampling uncertainty

The influence-function CI quantifies sampling uncertainty for the fitted statistical estimand.

Causal uncertainty

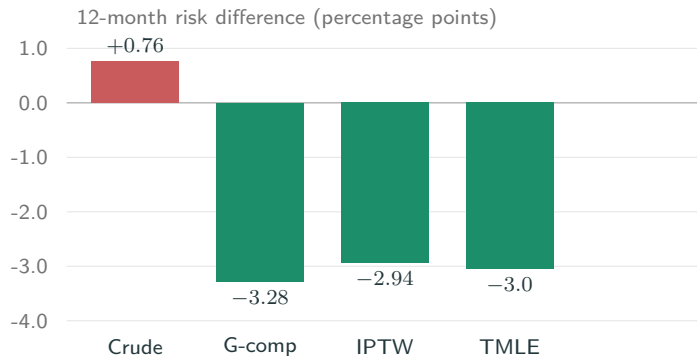
Separate analyses address hidden common causes, measurement, selection, treatment versions, and interference.

Descriptive tipping point

A +3.0-point shift moves the TMLE estimate to the null; this is not an E-value.

Adjustment reverses the sign

Result reveal



Synthetic teaching result — not clinical evidence.

Crude

+0.76 pp

TMLE

−3.0 pp

95% CI: −4.6 to −1.5 pp

What this shows

Adjusted analyses reverse the crude sign under the stipulated model. Agreement does not validate assumptions.

DEFENSIBLE

“In this synthetic exercise, adjustment reverses the crude association under the stipulated causal model.”

Supported claim

A statement about this workflow, this fixed dataset, and the assumptions explicitly recorded in the roadmap.

NOT DEFENSIBLE

“The vaccine prevents pneumonia in real patients.”

Still unknown

Clinical effectiveness, safety, use beyond this cohort, real EHR bias, and transmission effects.

Accept, correct, or reject?

Cite one Step 4 assumption and one limitation when revising the claim.

YOUR DECISION LOG

Choose one field from Steps 1–8.

1. What evidence makes the entry defensible?
2. What would make you revise it?
3. Which downstream decision changes if you revise it?

Pair protocol

Accept, correct, or reject the entry; compare reasoning, then export.

Participant kit

Participant workbook PDF
Complete DAGitty specification
Fixed synthetic dataset
Reproducible analysis script

Answer key after the activity

Completed eight-step roadmap
(Markdown)

Module F leaves us with:

- a specified target trial
- an explicit causal model
- an identification argument
- a statistical estimand
- a prespecified estimator and sensitivity plan

MODULE G HANDOFF

Use the same cohort and estimand to compare adjusted estimates, uncertainty, and the limits that remain.

Design first

Estimate second. Interpret last.